

1 OVERVIEW

This manual is applied to R-30*i*B Mate/ R-30*i*B Mate Plus controller (called R-30*i*B Mate/ R-30*i*B Mate Plus).

R-30*i*B Mate/ R-30*i*B Mate Plus has different controller depending on the required standards.

NRTL controller: To meet UL/CSA standard

CE controller : To meet Machinery Directive, Low voltage Directive, EMC Directive to cover the requirement of CE mark

This manual covers these controllers of the R-30*i*B Mate/ R-30*i*B Mate Plus.

The difference of NRTL and CE controller from the Basic controller is small as shown in Table 1 (ex. EMC parts, Breakers).

And the specific descriptions of CE and NRTL controller have notifications in this manual.

Table 1. Applied standards

	Functional Safety	EMC Standard	Robot Standard Electrical Standard	Requirement	Difference
Basic controller	ISO 13849-1 IEC 61508	-	-	-	-
NRTL controller		-	UL1740 CAN/CSA Z434 NFP A79	UL standard CSA standard •USA and Canada	•UL listed main breaker
CE controller		EN 55011 EN 61000-6-2 EN 61000-6-4	EN/ISO 10218-1 EN 60204-1	CE Marking •Europe	•Noise filter •EMC Cabinet •Shielded cable

This manual describes the maintenance and connection of R-30*i*B Mate/ R-30*i*B Mate Plus.

- Maintenance Part: Troubleshooting, and the setting, adjustment, and replacement of units
- Connection Part: Connection of R-30*i*B Mate/ R-30*i*B Mate Plus to the robot mechanical unit and peripheral devices, and installation of the controller



WARNING

Before you enter the robot working area, be sure to turn off the power to the controller or press the EMERGENCY STOP button on the operator's panel or teach pendant.

Otherwise, you could injure personnel or damage equipment.

2 CONFIGURATION

2.1 EXTERNAL VIEW OF THE CONTROLLER

The appearance and components might slightly differ depending on the controlled robot, application, and options used.

Fig.2.1 (a) to (b) show the view of R-30*i*B Mate/ R-30*i*B Mate Plus.

Fig.2.1 (c) to (g) show the construction of the R-30*i*B Mate/ R-30*i*B Mate Plus controller.

Fig.2.1 (h) to (j) show the external view of the operator's panel and teach pendant.

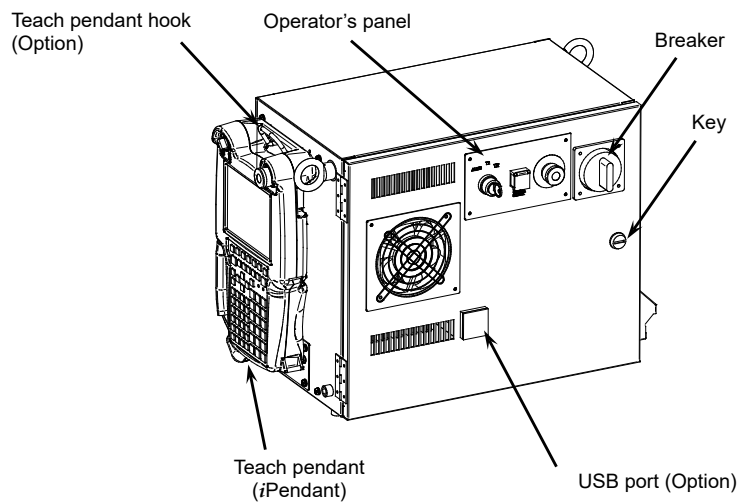


Fig.2.1 (a) External view of the R-30*i*B Mate/ R-30*i*B Mate Plus controller

NOTE

Be sure to lock the key.

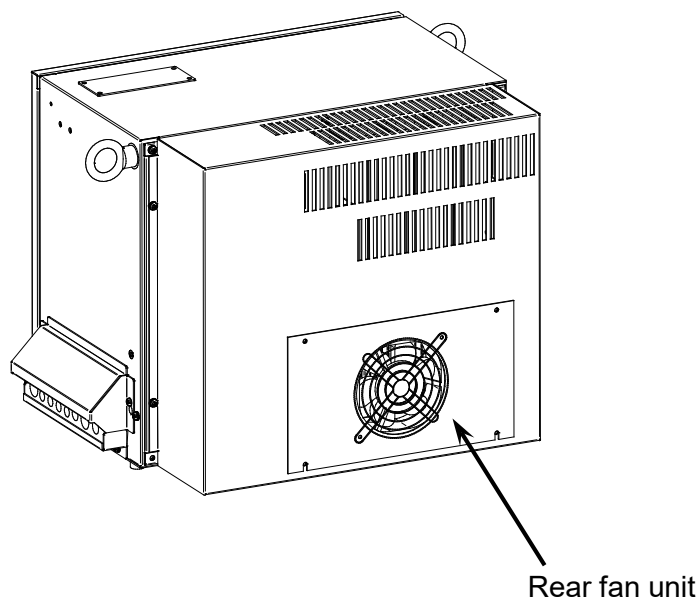


Fig.2.1 (b) External view of the R-30*i*B Mate/ R-30*i*B Mate Plus controller (Middle/Large size) (Rear)

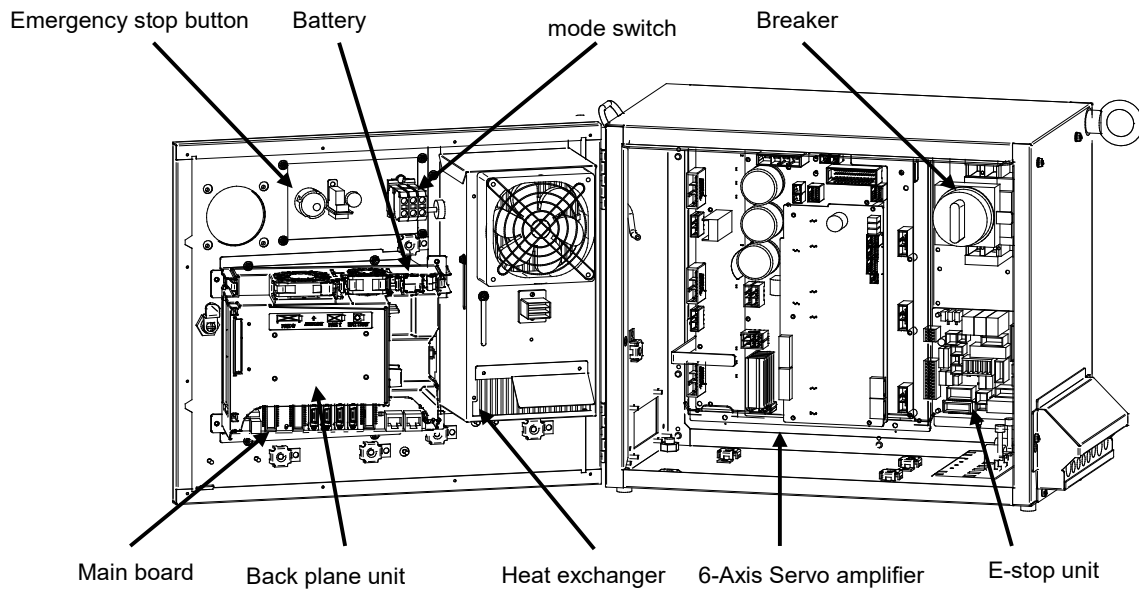


Fig.2.1 (c) R-30iB Mate/ R-30iB Mate Plus cabinet interior (Front-1)

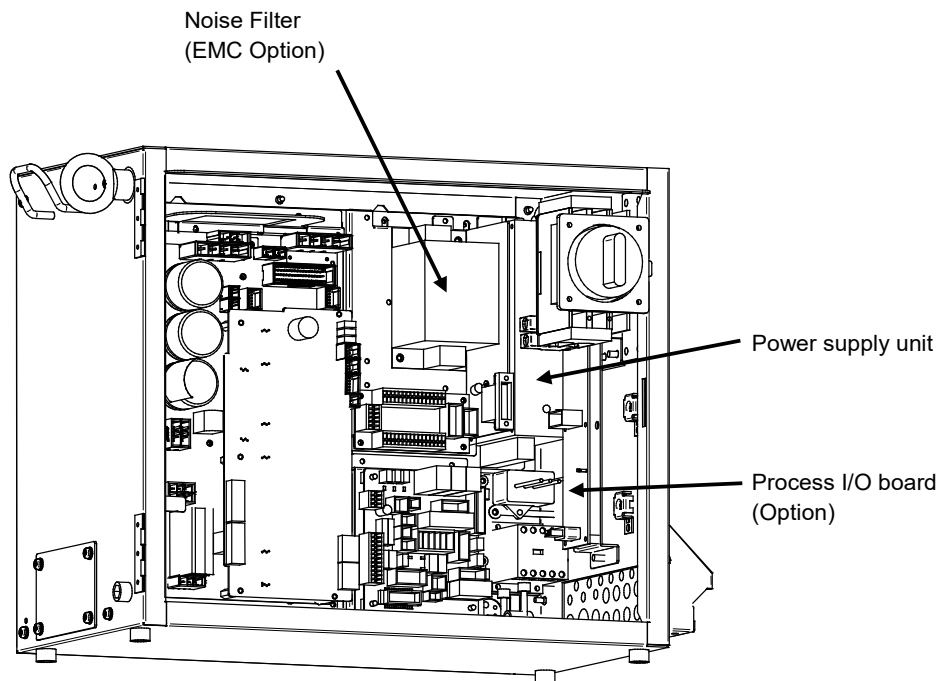


Fig.2.1 (d) R-30iB Mate/ R-30iB Mate Plus cabinet interior (Front-2)

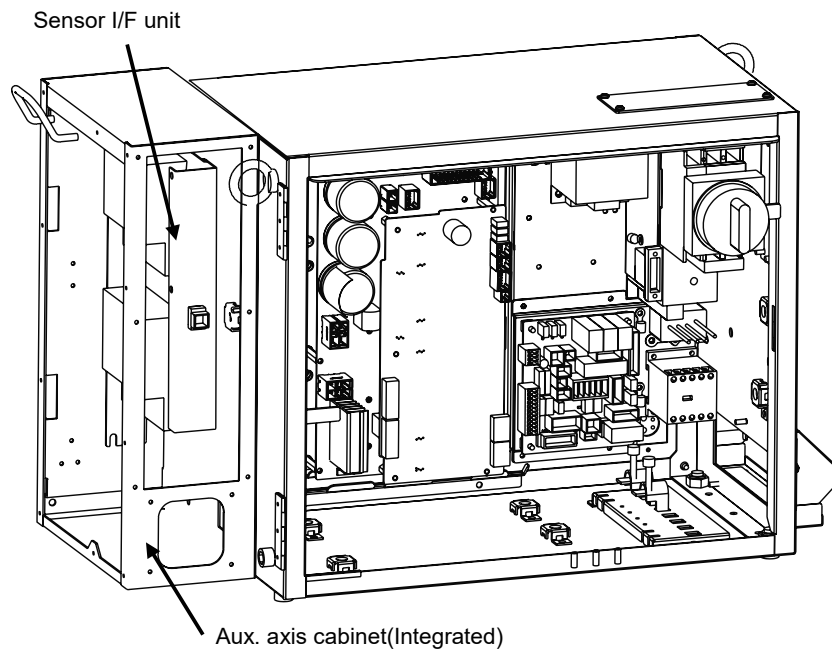
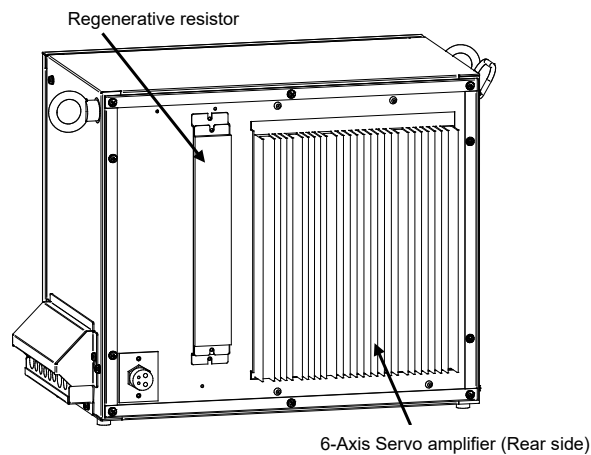
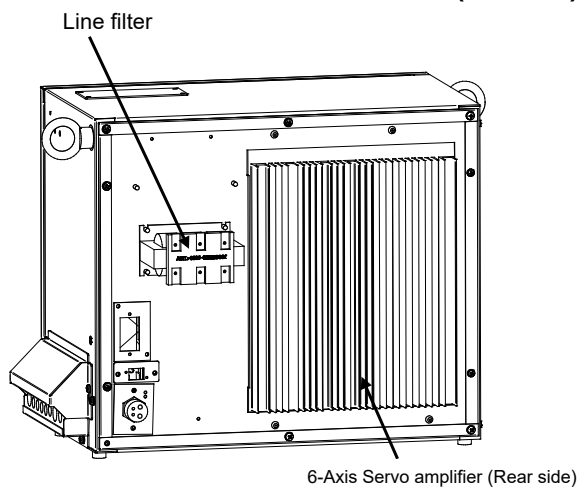


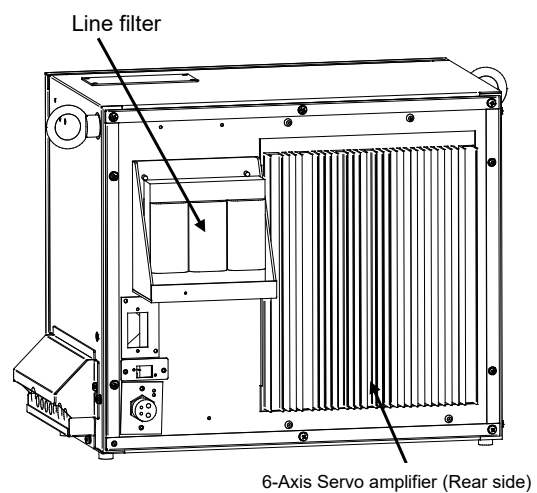
Fig.2.1 (e) R-30iB Mate/ R-30iB Mate Plus cabinet interior (Front-3) (CR-4iA, CR-7iA)



(Small size)



(Middle size)



(Large size)

Fig.2.1 (f) R-30iB Mate/ R-30iB Mate Plus cabinet interior (Rear)

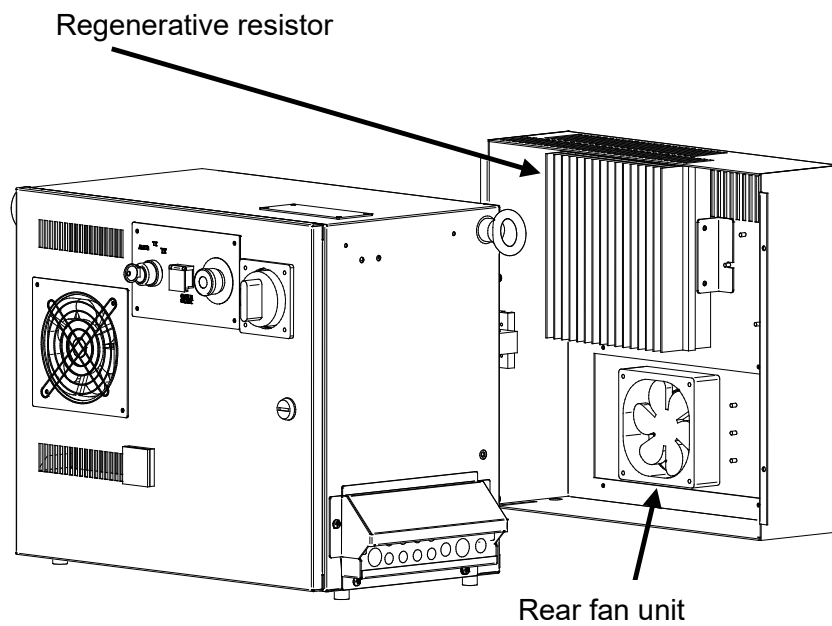


Fig.2.1 (g) R-30iB Mate/ R-30iB Mate Plus cabinet interior (Middle/Large size) (Rear)

NOTE

The number of fans on the fan unit (0,1,2) will vary depending on the robot type.

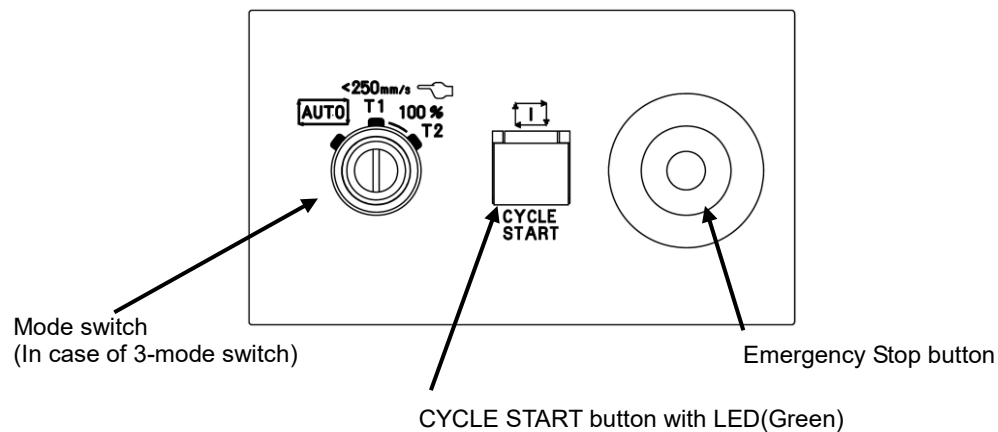


Fig.2.1 (h) R-30iB Mate/ R-30iB Mate Plus operator's panel

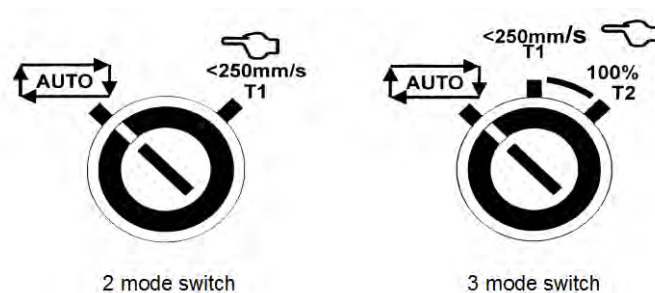


Fig.2.1 (i) Mode switch

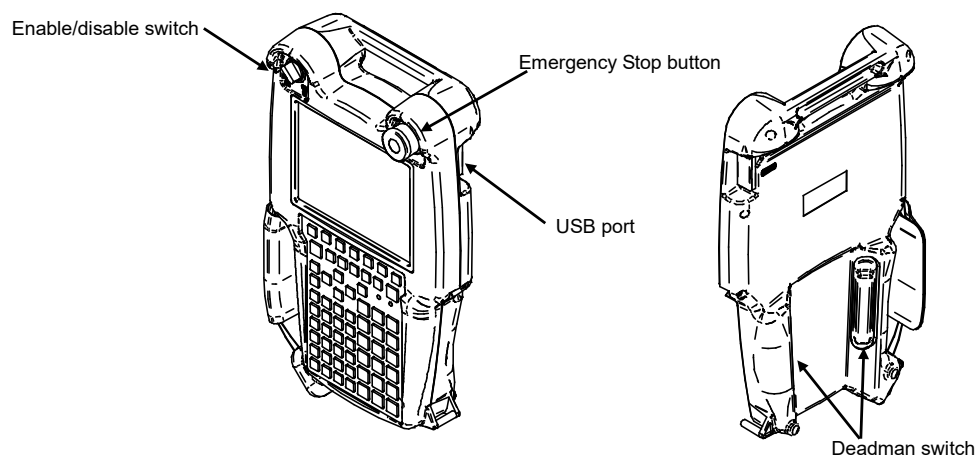


Fig.2.1 (j) Teach pendant (*iPendant*)

2.2 COMPONENT FUNCTIONS

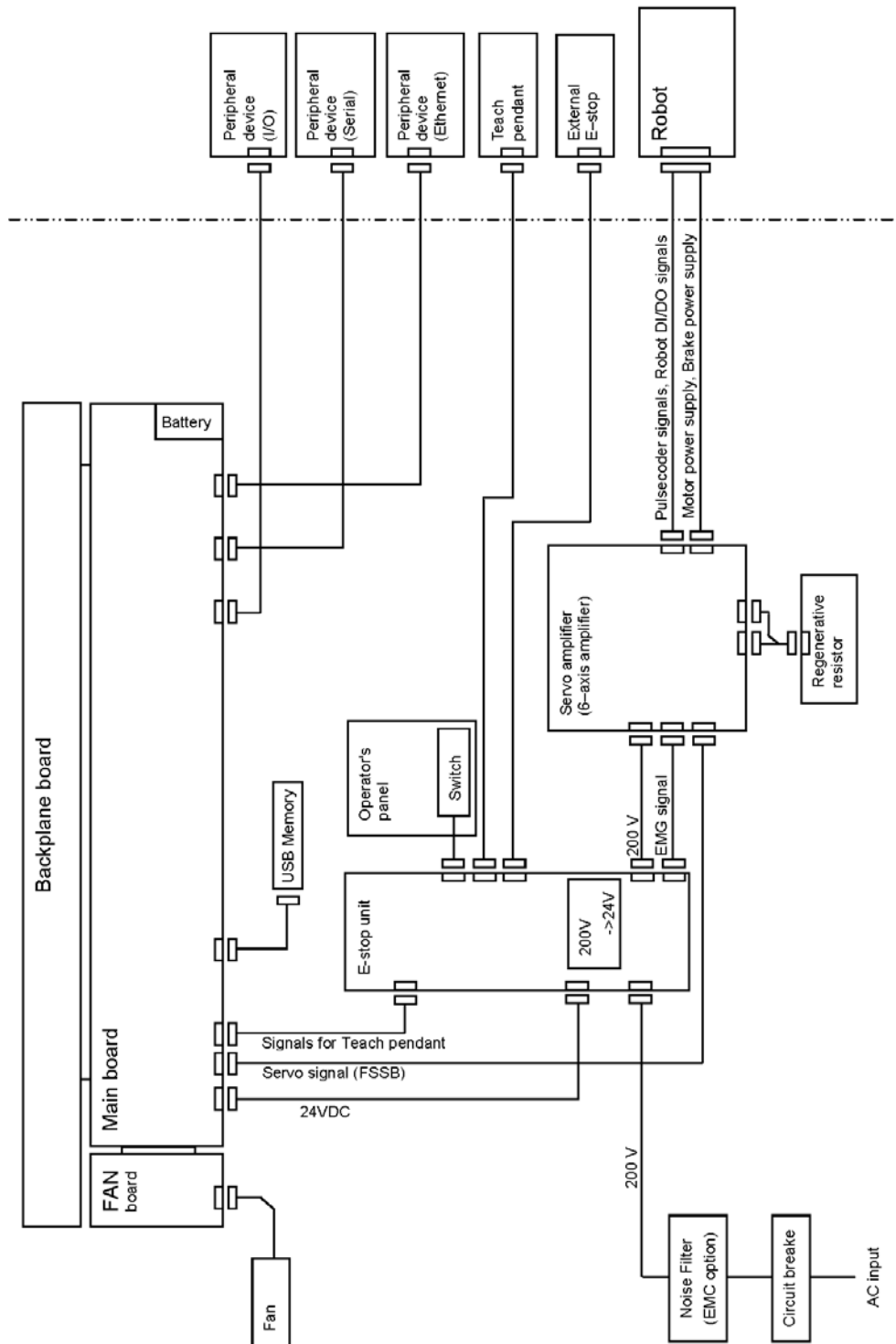


Fig.2.2 Block diagram of the R-30iB Mate/ R-30iB Mate Plus

- Main board
The main board contains a microprocessor, its peripheral circuits, memory, and operator's panel control circuit. The main CPU controls servo mechanism positioning.
- I/O printed circuit board
Various types of printed circuit boards are provided for applications including process I/O board. These are connected with FANUC I/O Link.
- E-stop unit
This unit controls the emergency stop system of the robot controller. It also has user interface terminals of safety relevant signals, external on/off signals etc.
- Power supply unit
The power supply unit converts the AC power to various levels of DC power.
- Backplane printed circuit board
The various control printed circuit boards are mounted on the backplane printed circuit board.
- Teach pendant
All operations including robot programming are performed with this unit. The controller status and data are indicated on the liquid-crystal display (LCD) on the pendant.
- 6-Axis Servo amplifier
The servo amplifier controls servomotor, Pulsecoder signal, brake control, overtravel and hand broken.
- Operator's panel
Buttons and LEDs on the operator's panel are used to start the robot and to indicate the robot status.
- Fan unit, heat exchanger
These components cool the inside of the controller.
- Circuit breaker
If the electric system in the controller malfunctions, or if abnormal input power causes high current in the system, the input power is connected to the circuit breaker to protect the equipment.
- Regenerative resistor
To discharge the counter electromotive force from the servomotor, connect a regenerative resistor to the servo amplifier.